

YI QING XU

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My research focuses on *translating* human objectives into signals robots can reliably act on. I develop *compositional and hierarchical abstractions as intermediate representations* between how people express intent and how robot policies, from specialized models to generalist VLAs, execute it, and design learning and inference methods that align robotic agents with human goals.

Keywords: Goal Specification for Robotics, Vision-Language-Action Models, Neuro-Symbolic Representation, Compositional Generative Models, Imitation Learning, IRL.

RESEARCH HIGHLIGHT

My work is organized around a common **abstraction-and-composition** principle: recover task structure implicit in human input, represent it explicitly, and use that structure to guide robot behavior. In “**SetItUp**” ([IJRR 2025](#)) and “**StackItUp**” ([CoRL 2025 Oral](#)), I map under-specified language and sketches into relational abstractions, then compose learned local models to generate feasible physical configurations.

My current work extends this idea to generalist robot policies. In “**The Human–VLA Communication Gap**” ([under review, ICRA 2027](#)), we characterize the mismatch between how people naturally express objectives and the executable structure VLAs require: humans increasingly compress task semantics and event structure as tasks grow, while VLAs are especially sensitive to missing semantics, undecomposed events, and how resolved commands are invoked. This motivates my current work on **structured and programmatic interfaces** that recover task structure before execution, together with **explicit concept memory**, **compositional skill chaining**, and **adaptive planning** ([APIVOT, NeurIPS 2026](#)) and **policy steering** ([VLS, CoRL 2026](#)).

Looking ahead, I aim to extend this framework in two directions. First, toward **mixed-modality goal specifications**, combining coarse, abstract instructions with precise but partial demonstrations to infer symbolic task skeletons and modular reward functions that can be composed and optimized jointly. Second, toward **interactive goal specification interfaces**, where robots engage with users via language, gaze, and motion to resolve ambiguity through active dialogue and inference, asking about the structure that is genuinely costly to guess rather than what the scene already disambiguates. Across both, the goal remains the same: generalist robots should adapt to how people actually communicate intent, rather than requiring people to adapt to the robot.

EDUCATION

National University of Singapore	Aug 02, 2020 - May 31, 2026
Ph.D in Computer Science, advised by Prof. David Hsu	GPA: 5.00

National University of Singapore	Aug 02, 2016 - Jun 30, 2020
B.A. in Computer Science	GPA: 4.81
B.A. in Applied Mathematics	GPA: 4.89

RESEARCH EXPERIENCE

Postdoctoral Scholar	Jul. 2026 – Present
<i>Stanford University, Computer Science, Prof. Jiajun Wu</i>	

Developing structured interfaces for generalist robot policies, with a focus on human–VLA communication, explicit concept memory, compositional skill chaining, and adaptive planning and policy steering.

Research Intern	Nov. 2025 – May 2026
<i>Allen Institute for Artificial Intelligence (AI2), Robotics (Seattle), Prof. Dieter Fox</i>	

Developed methods for task and goal specification for robot foundation models, studying how human objectives can be represented, grounded, and translated into robot-executable behavior.

Visiting Ph.D Student	Sep, 2023 - Feb, 2024
<i>CSAIL MIT, Prof. Leslie Kaelbling and Prof. Tomás Lozano-Pérez</i>	

Developed “*Set It Up!*”, a neuro-symbolic system that interprets and optimizes under-specified instructions for table-top arrangements using very few demonstrations. Published at RSS 2024.

SELECTED PUBLICATIONS / PREPRINTS

1. The Human-VLA Communication Gap: What Humans Leave Implicit and What Policies Cannot Recover [\[Link\]](#)
Yiqing Xu, Jiajun Wu
Under review, 2026.
2. APIVOT: Adaptive Planning with Interleaved Vision-Language Thoughts [\[Link\]](#)
Emily Jin, Joy Hsu, Yiqing Xu, Weiyu Liu, Nick Haber, Jiajun Wu
Advances in Neural Information Processing Systems (NeurIPS), 2026.
3. VLS: Steering Pretrained Robot Policies via Vision-Language Models [\[Link\]](#)
Shuo Liu, Ishneet Sukhvinder Singh, Yiqing Xu, Jiafei Duan, Ranjay Krishna
Conference on Robot Learning (CoRL), 2026.
4. “Set It Up”: Functional Object Arrangement with Compositional Generative Models [\[Link\]](#)
Yiqing Xu, Jiayuan Mao, Linfeng Li, Yilun Du, Tomás Lozano-Pérez, Leslie Kaelbling, David Hsu
International Journal of Robotics Research (IJRR), 2025.
5. “Stack It Up”: 3D Stable Structure Generation from 2D Hand-drawn Sketch [\[Link\]](#)
Yiqing Xu, Linfeng Li, Cunjun Yu, David Hsu
Conference on Robot Learning (CoRL), 2025.
6. On the Effective Horizon for Inverse Reinforcement Learning [\[Link\]](#)
Yiqing Xu, Finale Doshi-Velez, David Hsu
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025.
7. “Set It Up!”: Functional Object Arrangement with Compositional Generative Models [\[Link\]](#)
Yiqing Xu, Jiayuan Mao, Yilun Du, Tomás Lozano-Pérez, Leslie Kaelbling, David Hsu
Robotics: Science and Systems (RSS), 2024
8. Differentiable Particles for General-Purpose Deformable Object Manipulation [\[Link\]](#)
Siwei Chen, Yiqing Xu, Cunjun Yu, Linfeng Li, David Hsu
In submission.
9. Grounding Common-sense Objective for Tabletop Object Rearrangement [\[Link\]](#)
Yiqing Xu, David Hsu
Robotics: Science and Systems (RSS), 2023, Lang2Rob Workshop
10. Learning Reward for Physical Skills using Large Language Models [\[Link\]](#)
Yuwei Zeng, Yiqing Xu
Conference on Robot Learning (CoRL), 2023, the 2nd Lang2Rob Workshop
11. Benchmarking Deformable Object Manipulation with Differentiable Physics [\[Link\]](#)
Siwei Chen, Yiqing Xu*, Cunjun Yu*, Linfeng Li, David Hsu*
International Conference on Learning Representations (ICLR): **Notable Top 5%**, 2023
12. Receding Horizon Inverse Reinforcement Learning [\[Link\]](#)
Yiqing Xu, Wei Gao, David Hsu
Advances in Neural Information Processing Systems (NeurIPS), 2022
13. Coach: Cooperative robot teaching [\[Link\]](#)
Cunjun Yu, Yiqing Xu, Linfeng Li, David Hsu
Conference on Robot Learning (CoRL), 2022

PROFESSIONAL SERVICE

Workshop Organizer

- LEAP Workshop at CoRL 2026 November 2026
- Second LEAP Workshop at CoRL 2024 [\[Website\]](#) November 2024
- Task Specification Workshop at RSS 2024 [\[Website\]](#) July 2024

Reviewer: ICRA, RSS, AAMAS, AAAI, NeurIPS, ICLR, ICML

AWARDS

President's Graduate Fellowship, National University of Singapore (2020).

NUS Research Achievement Award, National University of Singapore (2022, 2025).

INVITED TALKS & PRESENTATIONS

Invited Talk, [Creativity in Assembly Workshop](#), IROS (2026).

Invited Talk, Stanford Vision and Learning Lab, Stanford University (2025).

Invited Talk, Personal Robotics Lab, University of Washington (2025).

Invited Talk, Robotics Team, Allen Institute for Artificial Intelligence (AI2) (2025).

Oral Presentation, Conference on Robot Learning (CoRL) (2025).

MENTORSHIP

Connie Jiang (Summer Intern) — Now B.S., Massachusetts Institute of Technology (2023).

Aishik Pyne (M.S., NUS) — Now Founder, Stealth Startup (2024).

Wanchen Li (Ph.D., University of Montpellier) — Ongoing Research Mentorship (2025–present).

Claire Yang (Ph.D., University of Washington) — Ongoing Research Mentorship (Jan 2026–present).

Emily Jin (Ph.D., Stanford University) — Ongoing Research Mentorship (Mar 2026–present).

Yize Liu (Ph.D., Stanford University) — Ongoing Research Mentorship (Jul 2026–present).

Jisu Han (Ph.D., Seoul National University) — Ongoing Research Mentorship (March 2026–present).